آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 16

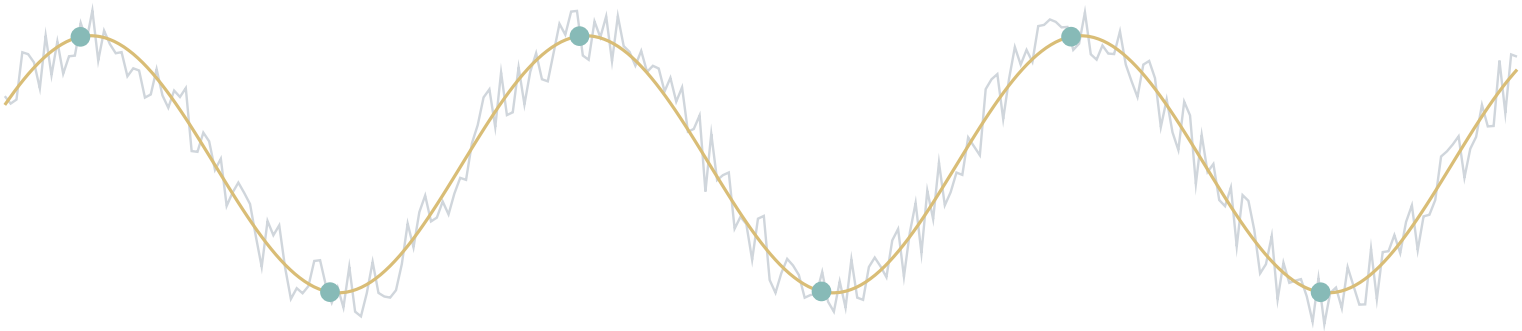
Engineering Statistical Applications

کاربردهای آماری مهندسی

Instructor: Amin Azari

Compiled by: Aidin Shekari

استاد: امین آذری github.com/AminAzari



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Learning Objectives

This capstone module integrates ALL previous modules into real Electrical Engineering applications. Students will be able to:

1. Apply probabilistic modeling to communication systems
2. Analyze signal processing problems statistically
3. Model electronic component variations
4. Assess power system reliability
5. Handle measurement uncertainty in control systems
6. Connect probability/statistics to machine learning foundations

16.1 Communications

16.1.1 BER Estimation and Confidence Intervals

Problem: Estimate the Bit Error Rate of a communication link.

Model: Each bit is a Bernoulli trial with error probability p (the BER).

- n bits transmitted, k errors observed
- $\hat{p} = \frac{k}{n}$ (MLE of BER)
- 95% CI: $\hat{p} \pm 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

16.1.2 AWGN Noise Modeling

Additive White Gaussian Noise is the fundamental noise model:

- **Distribution:** $N\left(0,\, \frac{N_0}{2}\right)$ per dimension
- **Why Gaussian?** CLT — thermal noise is sum of many electron contributions
- **Characterization:** Fully described by power spectral density N_0 or variance $\sigma^2 = N_0 B$ (B = bandwidth)

16.1.3 Rayleigh Fading Channel

Model: Channel coefficient $h = h_I + j \cdot h_Q$ where $h_I,\, h_Q \sim N(0,\, \sigma^2)$ independent.

- Envelope $|h| \sim$ Rayleigh(σ)
- Power $|h|^2 \sim$ Exponential $\left(\frac{1}{2\sigma^2}\right)$
- **Outage probability:** $P(|h|^2 < \gamma_{\text{th}}) = 1 - \exp\left(-\frac{\gamma_{\text{th}}}{2\sigma^2}\right)$

16.1.4 Rician Fading

When a line-of-sight (LOS) component exists:

- $h = h_{\text{LOS}} + h_{\text{scatter}}$
- $|h| \sim$ Rician$(\nu,\, \sigma)$ where $K = \frac{\nu^2}{2\sigma^2}$ is the K-factor

K-factor: $K = 0$ (Rayleigh) $\to$ $K = \infty$ (no fading, AWGN)

16.1.5 SNR Statistics

Instantaneous SNR: $\gamma = |h|^2 \times \frac{E_s}{N_0}$

For Rayleigh: $\gamma \sim$ Exponential$(\bar{\gamma})$ where $\bar{\gamma}$ = average SNR

Average BER over fading:

$$\bar{P}_e = \int_0^\infty P_e(\gamma) \cdot f_\gamma(\gamma)\, d\gamma$$

For BPSK over Rayleigh: $\bar{P}_e = \frac{1}{2}\left(1 - \sqrt{\frac{\bar{\gamma}}{1+\bar{\gamma}}}\right)$

16.2 Signal Processing

16.2.1 Measurement Noise Characterization

Problem: Characterize noise in measured signal $x(t) = s(t) + n(t)$.

Approach:

1. Collect noise-only samples (signal absent)
2. Estimate noise statistics: $\hat{\mu}_n$, $\hat{\sigma}_n^2$
3. Test for Gaussianity (histogram, QQ-plot)
4. Compute SNR: SNR $= \frac{P_{\text{signal}}}{\hat{\sigma}_n^2}$

16.2.2 Correlation-Based Detection

Problem: Detect a known signal in noise.

Method: Correlate received signal with known template. Under H_0 (noise only), the correlator output has known distribution $\rightarrow$ set threshold.

16.3 Electronics

16.3.1 Component Tolerance Modeling

Problem: Resistors have nominal value R_0 with manufacturing tolerance.

Model: $R \sim N(R_0, \sigma_R^2)$ where σ_R depends on tolerance class.

For $\pm 5\%$ tolerance (99.7% within bounds): $3\sigma = 0.05R_0 \rightarrow \sigma = 0.0167R_0$

16.3.2 Monte Carlo Circuit Analysis

Problem: A voltage divider uses R_1 and R_2. What is the distribution of output voltage?

$$V_{\text{out}} = V_{\text{in}} \times \frac{R_2}{R_1 + R_2}$$

With R_1, R_2 random: V_{out} is a random variable!

16.3.3 Measurement Uncertainty Propagation

If $Z = f(X, Y)$ and X, Y have small uncertainties:

$$\sigma_Z^2 \approx \left(\frac{\partial f}{\partial X}\right)^2 \sigma_X^2 + \left(\frac{\partial f}{\partial Y}\right)^2 \sigma_Y^2$$

Example: Power $P = \frac{V^2}{R}$. $\sigma_P^2 \approx \frac{2\text{V}}{R}^2 \sigma_V^2 + \frac{V^2}{R^2}^2 \sigma_R^2$

16.4 Power Systems

16.4.1 Load Uncertainty

Problem: Peak load is uncertain. Model: $L \sim N(\mu_L, \sigma_L^2)$

Design question: What capacity C ensures $P(\text{load} > \text{capacity}) < 0.01$?

$C = \mu_L + 2.326\sigma_L$ (for 1% exceedance probability)

16.4.2 System Reliability

Series system: All components must work.

$R_{\text{sys}} = R_1 \times R_2 \times \ldots \times R_n$ (for independent components)

Parallel system: At least one must work.

$R_{\text{sys}} = 1 - (1 - R_1)(1 - R_2)\ldots(1 - R_n)$

16.5 Control Systems

16.5.1 Sensor Noise Effects

Problem: A control system uses noisy sensor readings.

Measurement: $y(k) = x(k) + n(k)$ where $n \sim N(0, \sigma_n^2)$

Controller acts on $y(k) \rightarrow$ tracking error includes noise contribution.

Solution approaches:

- Averaging (low-pass filter): reduces noise by $\sqrt{n}$
- Kalman filter: optimal fusion of model prediction and measurement

16.5.2 Parameter Uncertainty

System model: $G(s) = \frac{K}{\tau s+1}$ where K and τ are uncertain.

Monte Carlo approach: sample K and τ from their distributions, simulate system response many times, compute statistics of output.

16.6 Machine Learning / AI Connections

16.6.1 Sampling and Data Collection

- Training data = sample from population
- Test data = independent sample for validation
- i.i.d. assumption underlies most ML theory

16.6.2 Estimation as Foundation of ML

- Linear regression = MLE under Gaussian noise assumption
- Neural network training = MLE (or MAP) for model parameters
- Loss function minimization = maximum likelihood principle

16.6.3 Probability Distributions in ML

- Generative models: learn $P(X)$ directly
- Gaussian Mixture Models: weighted sum of Gaussians
- Variational Autoencoders: latent variables with Gaussian priors

16.6.4 Bias-Variance Tradeoff

$$\text{Test Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise}$$

This is exactly the MSE decomposition from estimation theory (Module 13)!

- High bias: underfitting (too simple model)
- High variance: overfitting (too complex model)
- Optimal: balance between the two

16.6.5 Confidence and Uncertainty in Predictions

- Confidence intervals for model predictions
- Bayesian neural networks: posterior over weights $\rightarrow$ prediction uncertainty
- Calibration: predicted probabilities should match observed frequencies

16.7 Key Integration Points

Module	Application in EE
1. Probability	Event probabilities, fault diagnosis
2. Random Variables	Modeling uncertain quantities
3. Distributions	Choosing noise/fading/failure models
4. Moments	Signal power, noise power, SNR
5. Transformations	dB conversion, power from voltage
6 – 7. Joint/Functions	Signal+noise, system combinations
8. Conditional	Performance under different states
9. Correlation	MIMO channels, sensor arrays
10. Multiple RVs	Vector processing, beamforming
11. CLT	Why noise is Gaussian, averaging
12. Statistics	Estimating system parameters
13. Estimation	MLE for channel/noise parameters
14. Confidence	Quantifying measurement uncertainty
15. Hypothesis	Quality control, performance comparison

End of course modules. The modules conclude with engineering applications and a summary of key concepts.